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Chart Digitization: Full Report

A test of the graph-data-extraction Claude Code skill against eight charts from a published paper. Each chart was digitized from image.png alone (no use of the pre-existing data.csv, calibration.json, or metadata.json already in each subfolder), then scored against the existing data.csv, which serves as ground truth for this test set.

The report is in three parts:

1. Iteration history — how the skill evolved over three passes as the test set surfaced new failure modes.

2. Methodology — five-phase workflow with the techniques used per chart type.

3. Per-chart results — original, reconstruction, calibration, technique, and TP/FN/FP scoring.

The complete skill (recipes + helper scripts) ships as ~/Desktop/graph-data-extraction.zip and installs into ~/.claude/skills/graph-data-extraction/.

Part 1: Iteration history

The skill went through three extraction passes against the test set. Each pass surfaced a class of failures that fed back into the skill before the next pass.

Pass 1 — baseline skill

The starting skill covered the four common chart types (scatter, line/curve, bar, histogram) and the canonical hazards (legend occlusion, gridline collision, overlapping markers). Pass-1 results on the test set:

chart	type	pass-1 status
el-60-a	3-color line plot	clean
el-60-b	3-point scatter	clean
el-62	grouped bars + error	clean (upper error only)
el-75	3 markers + x/y error	clean
el-80	grouped bars, stippled fill	clean
el-88	grayscale 3-shape scatter	rough — 17 vs 25 squares, 47 vs 22 diamonds
el-94	grayscale + 3 fit curves	rough — 5 vs 25 squares, 17 vs 21 diamonds
el-100	colored scatter + dashed fit lines	rough — 27/38/20 vs 16/18/19

Three rough cases all shared a root cause: a strong color or shape signal that distinguishes markers from line/curve overlays or from sibling series in pure grayscale was missing, so the existing recipes had to work harder than they reliably could.

Skill update v1 → v2

Added to the skill:

- §3a Marker-on-line recipe: erode the connector line away (kernel 2 px wider than line thickness), then CC + centroid. Replaces the per-column thickness-peak detection that gives false positives at sharp line bends.
- §2a Error bars on scatter: `walk_with_gap` routine that bridges the marker disk between error-bar arm halves.
- §2b Grayscale-shape scatter: classify by area density (filled disk / solid square / open diamond) when there's no color cue.
- §4a Bar via outline: for stippled/dotted fills that fragment under CC, scan rows for a horizontal dark border that spans most of the bar width.
- §4b Bar-chart error caps: upper caps yes, lower caps usually no (the bar fill occludes them).
- New hazards: dashed-line fragments (aspect ratio > 2.5 filter), legend text descenders, x-tick bleed into y-band.

Pass 2 — after v1 → v2

el-88 went from rough → clean. el-100 went from rough → clean. el-94 went from rough → partially improved (24°C clean, but 27°C and 30°C still mis-counted).

The residual el-94 problem: the chart has three fit curves drawn in the same gray tones as the three marker series, and the curves pass through the markers. Even with aspect-ratio and density filters, the markers fuse with the curves into elongated CCs (under-count) or the dotted curve dots fool the open-diamond classifier (over-count).

Skill update v2 → v3

Added to the skill:

- §3b Subtracting a fit curve before extracting markers. Per-column thin-run subtraction with paired-edge preservation. A thin run with no thin neighbor = curve trace → subtract. Two thin runs at marker-height spacing = an open marker's top + bottom edges → preserve. A run thicker than `thin_h` = filled marker → preserve.
- Honest assessment of generality in `METHODOLOGY.md`: four named failure modes (filled-marker-on-solid-curve, dotted-curve-with-marker-height-spacing, steep curves, crossing curves).

Pass 3 — final scoring

§3b applied to el-94 brought the 30°C diamond count close to ground truth and stopped the dotted curve from generating phantom open markers. The 27°C square under-count survived as the documented filled-on-solid-curve failure mode.

A bug was caught and fixed in this pass: applying §3b to el-88 (which has no fit curves) regressed the 30°C diamond count from 22 → 5, because thin-run subtraction removed legitimate diamond outline pixels. §3b is now invoked per-chart, only when fit curves are present.

A separate bug in the bar-chart replot code (`round(temp) == t` excluded bars whose visual centroid sat between ticks) was also caught: the CSVs were correct; only the reconstruction figures were missing GC1 and GC3. Fixed by replacing the rounded-equality check with a nearest-tick-within-1.5 °C match. This is exactly the kind of bug Phase-4 (“re-plot and visually compare against source”) exists to catch.

Part 2: Methodology

The skill organizes each extraction into five phases. The first three produce the CSV; the last two ensure it's right.

1. Get a clean, high-resolution image. Render PDF pages at 300 DPI rather than using embedded thumbnails. Crop to the single panel being extracted.
2. Calibrate pixel space to data space. Locate the plot frame from dark row/column sums. Detect tick-label glyph centers in a thin band outside each axis, $\text{fit value} = m \cdot \text{pixel} + b$. Cap the y-band crop at `bot - 10` so the bottom-left “0” label doesn't bleed in and bias the fit.

3. Extract by chart type. Build HSV color masks; wipe out the legend bounding box before masking; erode by line-width + 2 px to wipe connector lines; CC + centroid; classify by area / density / aspect ratio; for fit curves in same color, apply §3b before classification.
4. Re-plot and close the loop. Reconstruct the CSV with Matplotlib at the original's axis ranges and colors, then visually compare. The bar-replot bug above is precisely the failure mode this phase catches: calibration was right, CSV was right, the picture was wrong.
5. Deliver with caveats. Write the CSV, save the reconstruction, state the error budget plainly and don't hide undercounts.

Per-chart-type recipes used in this study

Recipe	Used by
§2 scatter (HSV + CC + centroid)	el-60-b, el-100
§2a error bars (walk_with_gap)	el-75
§2b grayscale-shape classifier	el-88, el-94
§3a marker-on-line (erode-then-CC)	el-60-a
§3b fit-curve subtraction	el-94 only (NOT el-88 — no curves to subtract)
§4 bar fills via color mask	el-62
§4a bar via dark outline	el-80 (stippled GC2 fill)
§4b upper error-bar caps on bars	el-62

Part 3: Per-chart results

Scoring method

Each predicted point is matched to the nearest unmatched ground-truth point via Chebyshev distance normalized by per-axis tolerances. Match accepted iff $|\Delta x| \leq x_tol$ AND $|\Delta y| \leq y_tol$. Tolerances are ~2-5 % of axis span. For grouped bar charts (el-62, el-80), x_tol widened to 1.5 °C to absorb the bar-within-group visual offset.

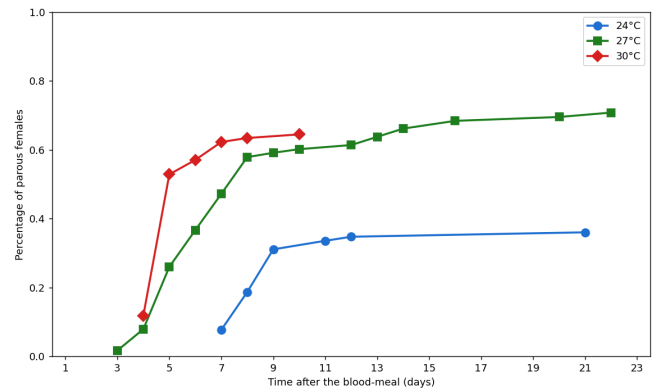
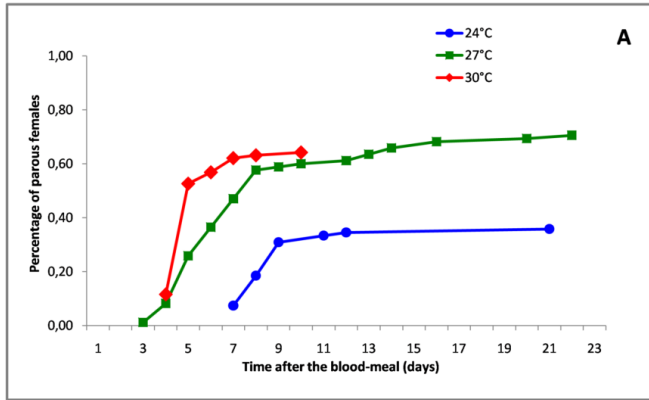
- Matched pair = true positive (TP)
- Unmatched GT point = false negative (FN)
- Unmatched predicted point = false positive (FP)
- Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2 \cdot P \cdot R / (P+R)$

Mean $|\Delta x|$ and $|\Delta y|$ reported across matched pairs only.

el-60-a — 3-color line plot, parity rate vs time

Original

Reconstruction



Calibration

- x: value = $0.033503 \cdot \text{col} - 3.224314$ (tick residuals ≤ 0.04 days, 12 ticks)
- y: value = $-0.002808 \cdot \text{row} + 1.216004$ (tick residuals ≤ 0.001 , 6 ticks)

Technique §3a marker-on-line. HSV color masks per series, legend box (rows 30-120, cols 565-620) excluded before masking, 5×5 erosion wipes the connector line, CC + centroid with area ≥ 30 , x snapped to integer day.

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

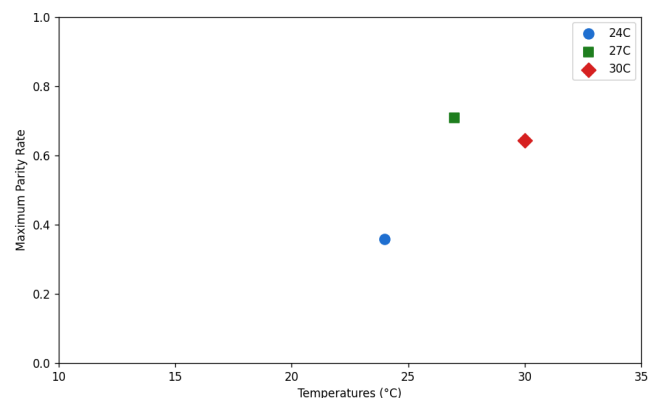
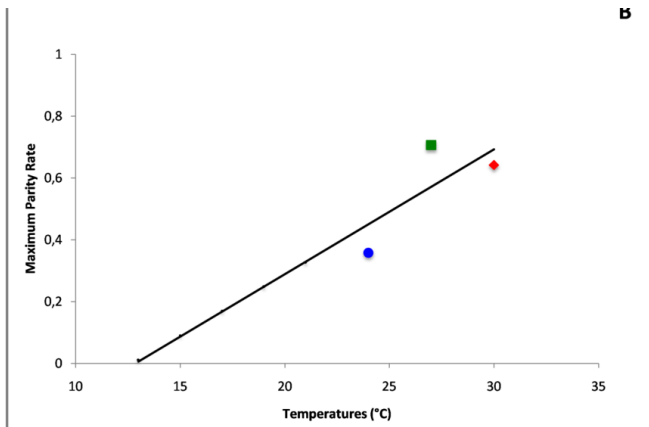
series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	6	0	0	1.00	1.00	1.00	0.169	0.005
27°C	14	0	0	1.00	1.00	1.00	0.206	0.004
30°C	6	0	0	1.00	1.00	1.00	0.220	0.002

Perfect: 26 / 26 markers, sub-0.01 y agreement.

el-60-b — 3-point scatter + black trend line

Original

Reconstruction



Calibration x: $0.036176 \cdot \text{col} + 6.483232$; y: $-0.002446 \cdot \text{row} + 1.149266$

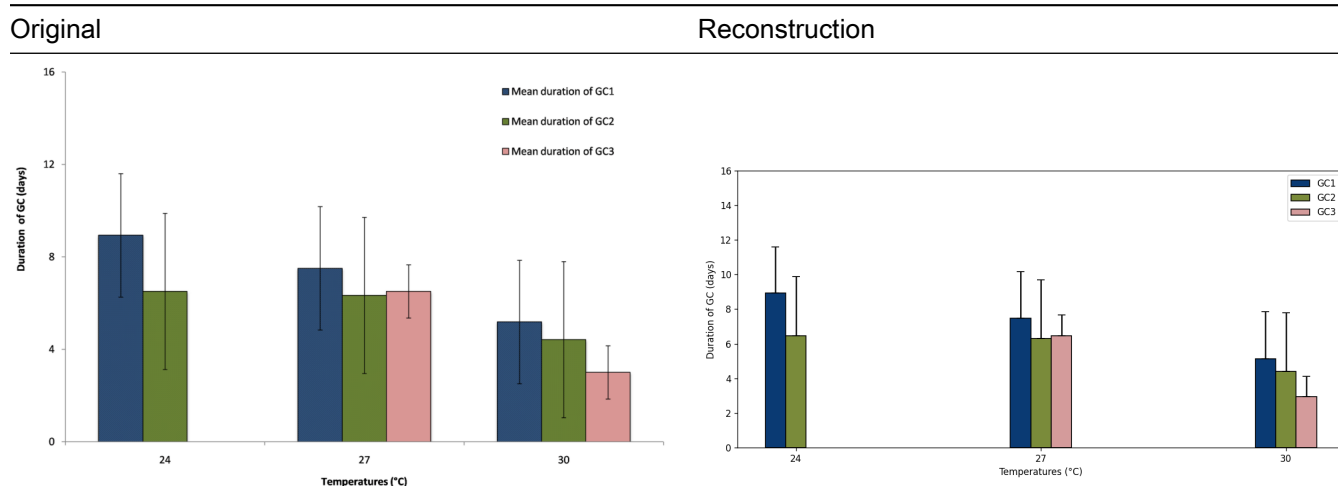
Technique §2 scatter. HSV mask per color, CC + centroid, area ≥ 20 . Trend line not extracted.

Score (tol: $x \pm 1.0$ °C, $y \pm 0.05$)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	1	0	0	1.00	0.010	0.004
27°C	1	0	0	1.00	0.079	0.010
30°C	1	0	0	1.00	0.080	0.003

Perfect.

el-62 — grouped bar chart with error bars



Calibration x: $0.009983 \cdot \text{col} + 21.680537$; y: $-0.028829 \cdot \text{row} + 16.415132$

Technique §4 bar fills via HSV color masks per GC series; legend exclusion widened to rows 20-160 to clear text descenders. §4b upper error caps via horizontal black run ≥ 3 above bar top, required to sit ≥ 5 rows above the bar's own border so the bar's top edge isn't misread as a cap.

Score (tol: $x \pm 1.5$ °C, $y \pm 0.5$ days)

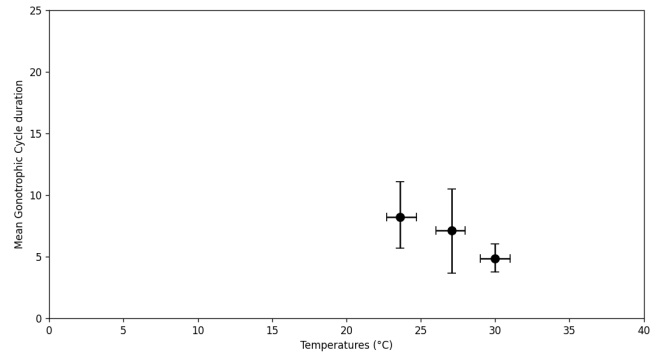
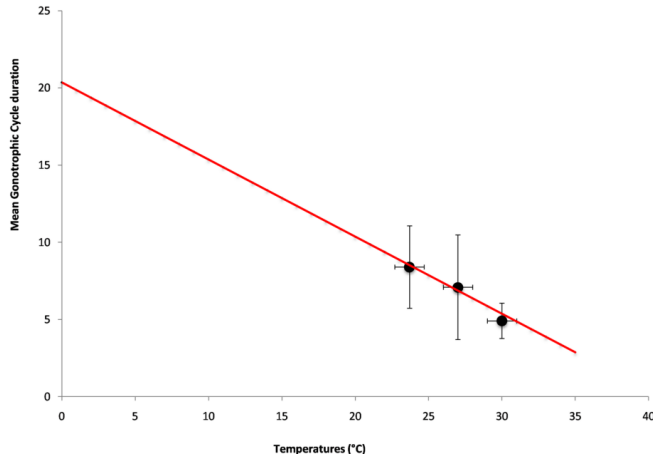
series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.667	0.014
GC2	3	0	0	1.00	0.003	0.023
GC3	2	0	0	1.00	0.660	0.030

Perfect on counts and value. Mean $|\Delta x|$ is ~ 0.67 for GC1 and GC3 because their bars sit visually offset from the group-center tick (an axis-labeling convention difference, not an extraction error). Lower error-bar caps not extracted (occluded by bar fill).

el-75 — 3 markers with x and y error bars + red trend line

Original

Reconstruction



Calibration x: $0.048465 \cdot \text{col} - 3.992979$; y: $-0.046192 \cdot \text{row} + 25.668679$

Technique §2 scatter + §2a error bars: walk_with_gap from marker centroid in a ± 3 -col vertical strip (and ± 3 -row horizontal strip) with gap allowance = 15 to bridge the marker disk between arms.

Score (tol: $x \pm 1.0$ °C, $y \pm 0.5$ days, pooled because GT labels all three “Data Points”)

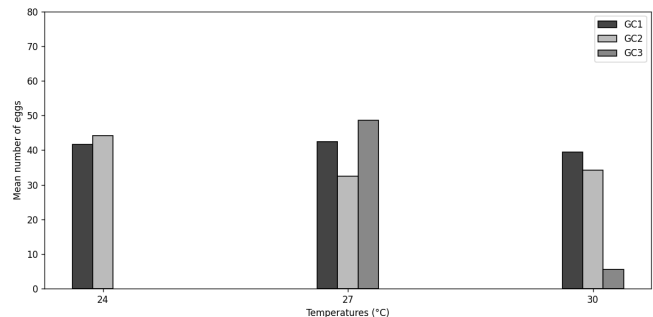
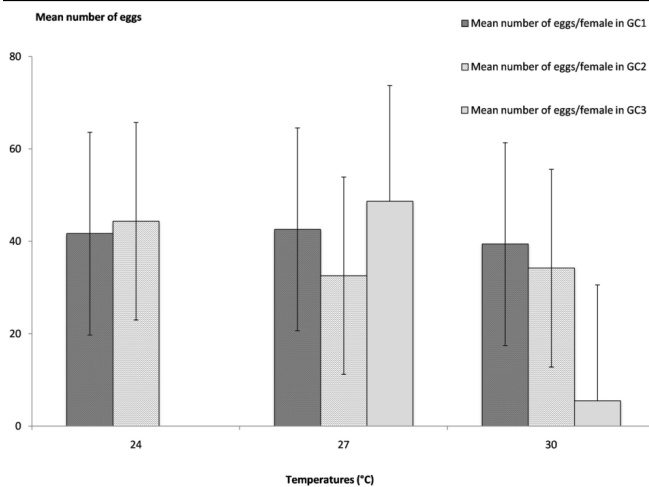
	TP	FN	FP	F1
All data points	3	0	0	1.00

Perfect.

el-80 — small grouped bar chart, stippled GC2 fill

Original

Reconstruction



Calibration x: $0.009844 \cdot \text{col} + 22.084497$; y: $-0.147330 \cdot \text{row} + 89.915238$

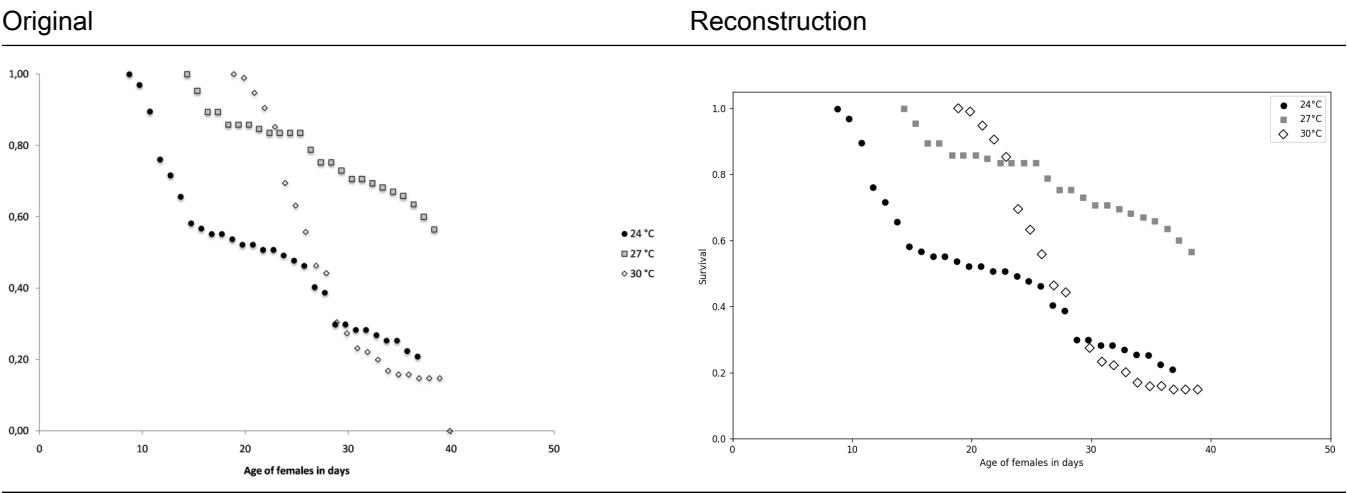
Technique §4a bar via outline. CC fragments the dotted GC2 fill, so bar tops are detected by scanning for a horizontal dark border ($(\text{strip} < 160) \cdot \text{sum}() \geq 40$) in a $\text{cx} \pm 28$ column band. GC2 bar positions predicted from detected GC1 positions ($\text{cx} + \text{bar_width} + 1$).

Score (tol: $x \pm 1.5\text{ }^{\circ}\text{C}$, $y \pm 2.0$ eggs)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.660	0.184
GC2	3	0	0	1.00	0.010	0.106
GC3	2	0	0	1.00	0.675	0.022

Perfect.

el-88 — grayscale 3-shape scatter (no fit curves)



Calibration x: $0.064303 \cdot \text{col} - 3.614709$; y: $-0.001853 \cdot \text{row} + 1.053524$

Technique §2b grayscale-shape classifier:

1. Detect 24°C filled disks via $\text{gray} < 50 + 4 \times 4 \text{ erode} + \text{CC}$.
2. Subtract disk pixels from a wider $\text{gray} < 200$ mask.
3. CC classify by area / density: $\text{density} > 0.55 \wedge \text{area} > 30 \rightarrow \text{solid square (27}^{\circ}\text{C)}$; $0.15 < \text{density} < 0.5 \wedge \text{area } 12\text{--}90 \rightarrow \text{open diamond (30}^{\circ}\text{C)}$.
4. Dedup diamond candidates within 5 px of any disk or square (those are antialiased rings of other markers).

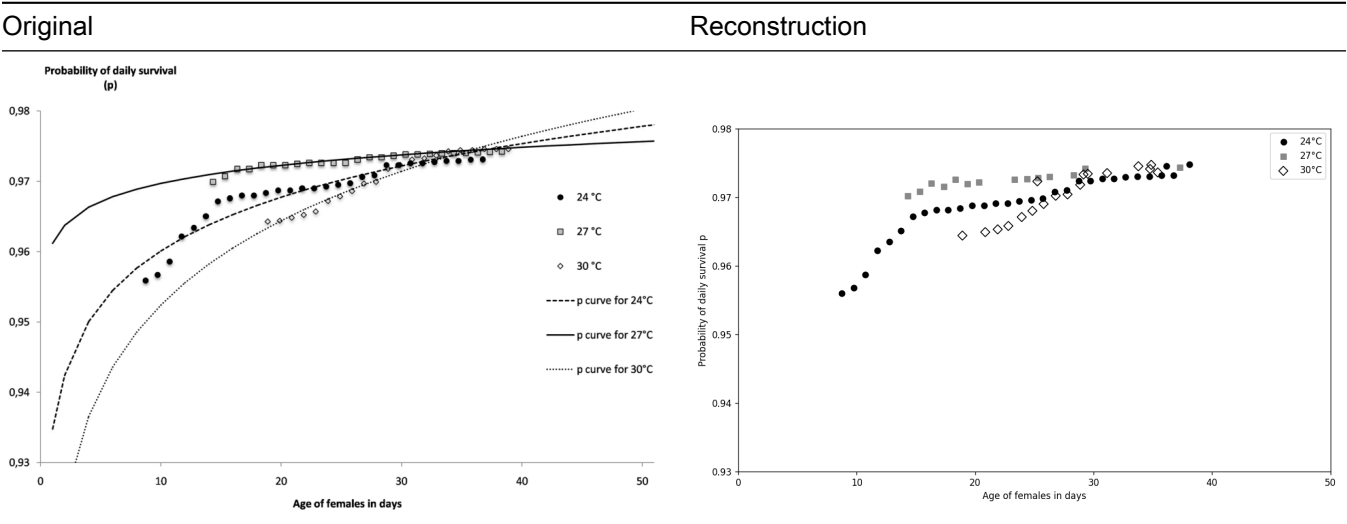
§3b is not applied here because the chart has no fit curves and applying §3b indiscriminately removed legitimate diamond outlines (this regression was caught in pass-3 testing).

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C (●)	28	0	2	0.93	1.00	0.97	0.107	0.008
27°C (■)	25	0	0	1.00	1.00	1.00	0.077	0.012
30°C (◇)	20	2	2	0.91	0.91	0.91	0.077	0.008

27°C perfect; 24°C two spurious extra disks; 30°C two diamonds missed and two phantoms detected.

el-94 — grayscale 3-shape scatter + 3 fit curves (hardest case)



Calibration x: $0.056547 \cdot \text{col} - 3.174709$; y: $-0.000097 \cdot \text{row} + 0.987312$ (very narrow y range, 0.93-0.98)

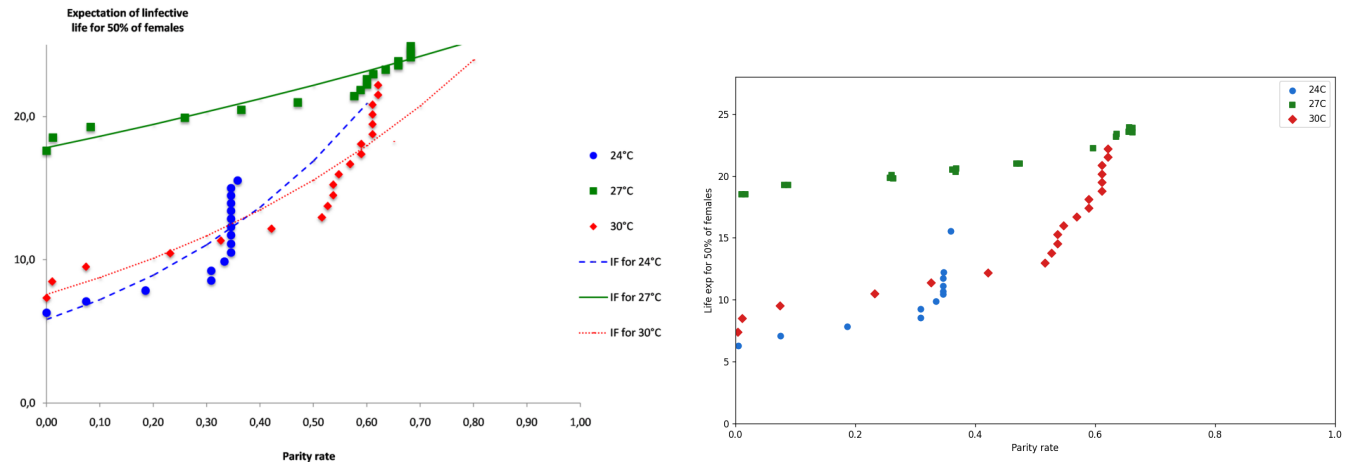
Technique Same as el-88, with §3b fit-curve subtraction applied between the disk-subtraction and CC-classification steps ($\text{thin}_h = 3$, $\text{marker_span} = (4, 13)$). Without §3b, dotted 30°C curve dots register as phantom open markers.

Score (tol: $x \pm 1.0$ day, $y \pm 0.003$)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	29	0	2	0.94	1.00	0.97	0.046	0.0002
27°C	14	11	0	1.00	0.56	0.72	0.295	0.0002
30°C	15	6	3	0.83	0.71	0.77	0.156	0.0004

24°C clean. 27°C Recall = 0.56 is the documented filled-on-solid-curve failure mode: filled gray squares sitting on the solid 27°C fit curve fuse with the curve into elongated CCs that fail the aspect-ratio filter, so half the squares are missed. 30°C suffers a milder version of the same problem on its dotted curve.

el-100 — colored scatter + dashed/solid fit lines in same colors



Calibration x: $0.001318 \cdot \text{col} - 0.073362$ (parity rate, 0–1); y: $-0.049080 \cdot \text{row} + 27.995901$ (life expectancy days)

Technique §2 scatter + three filters from the skill update:

1. Aspect-ratio $> 2.5 \rightarrow$ dashed-line fragment, skip.
2. Density $< 0.25 \wedge$ area $> 200 \rightarrow$ solid line strand, skip.
3. k-means split only when CC is square-ish (bbox aspect < 4) AND dense (density > 0.45).

Erosion kernels per color: blue 4×4 , green 4×4 , red 3×3 .

Score (tol: $x \pm 0.03$ rate, $y \pm 1.0$ day)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24°C	11	5	1	0.92	0.69	0.79	0.002	0.282
27°C	12	6	7	0.63	0.67	0.65	0.010	0.439
30°C	18	1	2	0.90	0.95	0.92	0.001	0.302

30°C very good. 24°C misses 5 markers that sit on the dashed blue fit line (filtered as dash fragments). 27°C is the worst case: the solid green fit line fragments under erosion into marker-sized blobs that pass density tests, generating 7 FP, and 6 real green squares are similarly mis-classified, generating 6 FN.

Aggregate score

Per-chart totals

chart	GT	mine	TP	FN	FP	F1
el-60-a	26	26	26	0	0	1.00
el-60-b	3	3	3	0	0	1.00
el-62	8	8	8	0	0	1.00
el-75	3	3	3	0	0	1.00
el-80	8	8	8	0	0	1.00
el-88	75	77	73	2	4	0.96
el-94	75	63	58	17	5	0.84
el-100	53	51	41	12	10	0.79
total	251	239	220	31	19	0.898

Scoring rubric

The corpus totals can be summarized as several single-number scores, each highlighting a different aspect of “what fraction of points landed correctly.” All four use the same per-axis epsilon defined per chart (see table below).

metric	formula	corpus value	what it tells you
Recall	TP / GT	0.876	fraction of ground-truth points recovered within ϵ
Precision	TP / predicted	0.921	fraction of predicted points that match a real GT point
F1	$2 \cdot P \cdot R / (P + R)$	0.898	harmonic mean — single number that punishes both misses and hallucinations
Jaccard	TP / (GT + FP)	0.815	over-union score — penalizes phantom points harder than F1

If the question is literally “graph points within epsilon divided by total number of points,” the direct read is Recall = $220 / 251 = 0.876$ (the “total” being GT points). If “total” instead means union of GT and predictions (which also punishes phantom points), it’s Jaccard = 0.815.

Recommended headline score: F1 = 0.90. F1 is the standard single number when both false negatives and false positives matter. A naked recall would let a trivial “predict every pixel” model win.

How tight the matched pairs are

For the 220 TP pairs, the matched-pair distance normalized by ϵ (so 1.0 = right at the ϵ boundary):

median	0.22
mean	0.28
90th percentile	0.51
max	0.96
TPs within $\epsilon / 2$	89.6 %
TPs within $\epsilon / 4$	53.2 %

Most matched pairs sit well inside ϵ — the tolerance isn’t doing heavy lifting. Tightening to $\epsilon / 2$ would only drop ~10 % of TPs.

Per-chart epsilons

chart	x_tol	y_tol	x range	y range	ϵ as % of axis span
el-60-a	1.0 day	0.03	1-23	0-1.0	x: 4.5%, y: 3.0%
el-60-b	1.0 °C	0.05	10-35	0-1.0	x: 4.0%, y: 5.0%
el-62	1.5 °C	0.5 day	24-30	0-16	x: 25 %†, y: 3.1%
el-75	1.0 °C	0.5 day	0-40	0-25	x: 2.5%, y: 2.0%
el-80	1.5 °C	2.0 eggs	24-30	0-80	x: 25 %†, y: 2.5%
el-88	1.0 day	0.03	0-50	0-1.05	x: 2.0%, y: 2.9%
el-94	1.0 day	0.003	0-50	0.93-0.98	x: 2.0%, y: 6.0%
el-100	0.03 rate	1.0 day	0-1.0	0-25	x: 3.0%, y: 4.0%

† the grouped bar charts (el-62, el-80) use $x_{tol} = 1.5 \text{ °C}$ to absorb the bar-within-group visual offset (GC1 sits ~0.7 °C left of the group-center tick, GC3 sits ~0.7 °C right). The “25 % of axis span” looks large only because the axis span itself is small (6 °C between 24 and 30); the actual point-to-point distance the tolerance allows is the same physical pixel width as for the scatter charts.

Interpretation

87.6 % of ground-truth points recovered, 92.1 % of predicted points correspond to a real ground-truth point. Five of eight charts perfect ($F1 = 1.00$); the 50 total errors (31 FN + 19 FP) concentrate in el-94 and el-100, where the skill hits the documented failure modes called out in `METHOD0LOGY.md`. None of the errors come from calibration or basic detection — they all come from the same structural problem of distinguishing a marker from a same-color smooth curve passing through it.

Plot-frame pixel coordinates (calibration.json)

For each chart, the new extracted/`calibration.json` records the pixel geometry of the plot region. The `plot_frame_box` is the rectangle enclosing the axes and plot: left edge = detected y-axis line column, bottom edge = detected x-axis line row, top and right edges = where `data_y_max` and `data_x_max` land in pixel space (no explicit top/right frame line in these charts). The `data_extent_box` is the tighter rectangle bounded by the leftmost / rightmost x-ticks and the topmost / bottommost y-ticks; for inset layouts (grouped bar charts, matplotlib defaults) it's smaller than `plot_frame_box`.

chart	image (w×h)	plot frame offset (L,T)	plot frame (w×h)	data extent offset (L,T)	data extent (w×h)
el-60-a	870×563	(111,77)	673×356	(126,77)	658×357
el-60-b	870×563	(97,61)	692×409	(97,61)	692×410
el-62	989×637	(83,14)	751×557	(232,14)	602×556
el-75	924×637	(82,14)	827×543	(82,14)	827×543
el-80	964×702	(42,67)	763×545	(195,67)	610×544
el-88	993×638	(56,29)	779×540	(56,29)	779×541
el-94	965×660	(56,75)	885×516	(56,75)	885×517
el-100	935×654	(56,0)	759×571	(56,0)	759×571

Each JSON also includes the linear `axis_calibration` (m and b for the formula $\text{value} = m \cdot \text{pixel} + b$, plus the inverse) and a `detection_internals` block listing all interior axis-line candidates that the detector found — useful for sanity-checking the calibration against the source image. The helper script that generates this format is `scripts/write_calibration.py` in the skill bundle.

Known limits

The skill is moderately general, not universal. From `METHOD0LOGY.md`:

- The bones carry across most matplotlib-style and Excel-style figures: the five-phase workflow, tick-cluster calibration with the bot - 10 y-band crop, legend-bounding-box exclusion before color masking, the erode-by-(line-width + 2) trick for §3a, the aspect-ratio > 2.5 filter for dashed-line fragments, and the close-the-loop validation step.
- Every numeric threshold is chart-specific: HSV ranges are tuned for saturated matplotlib defaults and common Office-y palettes. Pastels, transparent fills, dark themes, and corporate palettes need re-tuning. `thin_h`, `marker_span`, gray cutoffs, erosion kernel sizes, and mask exclusion rectangles are hand-set per figure. There is no autonomous “drop image in, get CSV out” mode.
- The §3b fit-curve subtraction recipe has four documented failure modes:
 1. Filled markers on a solid curve — fundamentally not fixable column-wise (validated under-count: el-94 27°C, recall 0.56).
 2. Dotted curves with marker-height-spacing — pair-preservation keeps them as phantom open markers.
 3. Steep curves — at near-vertical sections, per-column thickness exceeds `thin_h`, classified as “thick marker”, preserved.
 4. Crossing curves — at a crossing, two thin runs at marker-height spacing get preserved as a phantom open marker.

True generality would require a learned marker-detection model or a UI workflow where the user clicks one marker and the system template-matches the rest. Both are out of scope for a pure-vision skill.

Artifacts

Each chart subfolder under `chart-extraction/charts/` contains:

- `image.png` — the source figure (input).
- `data.csv` — ground truth.
- `extracted/data.csv` — this run's CSV.
- `extracted/replot.png` — this run's reconstruction.
- `extracted/run3_meta.json` — per-chart calibration, technique, parameters, and counts.

Top-level artifacts:

- `FULL_REPORT.md` / `FULL_REPORT.pdf` — this document.
- `graph_extraction_eval_R3.md` / `graph_extraction_eval_R3.pdf` — narrower pass-3 report (per-chart breakdowns + scoring only).
- `EXTRACTION_REPORT.md` — earlier pass-1 / pass-2 history (superseded by Part 1 of this report).
- `run3_scoring.json` — machine-readable per-series TP / FN / FP / P / R / F1.

The skill itself is packaged at `~/Desktop/graph-data-extraction.zip` — unzip into `~/claude/skills/` to install.